Additional file 2

Supplementary Table S1: Study characteristics

Table S1 shows detailed characteristics of the 54 included studies. IRB: institutional review board; NR: not reported.

First author	Month of publication	Country	Study design	IRB approval	No of patients
Al-Samkari	July 2020	USA	retrospective, multicenter	yes	144
Aleva	August 2020	Netherlands	retrospective; single-center	inf. consent waived	50
Avruscio	Septemb 2020	Italy	prospective; observational; single-center	yes	41
Azoulay	August 2020	France	retrospective; multicenter	inf. consent waived	376
Beun	April 2020	Netherlands	retrospective, single-center	inf. consent waived	75
Bilaloglu	August 2020	USA	retrospective; multicenter	inf. consent waived	829
Chen	June 2020	China	retrospective, single-center	inf. consent waived	88
Chistolini	July 2020	Italy	prospective; observational; single-center	yes	27
Criel	July 2020	Belgium	prospective, observational, single-center	NR	30
Cui	May 2020	China	retrospective, single-center	yes	81
Desborough	May 2020	UK	retrospective, single-center	yes	66
Devreese	July 2020	Belgium	prospective, observational, single-center	yes	31
Dujardin	September 2020	Netherlands	retrospective; single-center	no	127
Elbadawi	September 2020	USA	retrospective; multicenter	yes	296
Fraissé	June 2020	France	retrospective, single-center	yes	92
Gibson	September 2020	USA	prospective; observational; single-center	NR	72
Grandmaison	June 2020	Switzerland	prospective, observational, single-center	NR	29
Hekimian	June 2020	France	retrospective, single-center	informed consent	51
Helms	May 2020	France	prospective, observational, multicenter, control	yes	150
Hippensteel	June 2020	USA	retrospective, single-center	yes	91
Kapoor	October 2020	USA	prospective; observational; multicenter	yes	107
Klok	April 2020	Netherlands	retrospective, multicenter	inf. consent waived	184
Lemos	September 2020	Brazil	prospective; interventional; single-center	informed consent	20
Lendorf	August 2020	Denmark	retrospective; single-center	inf. consent waived	20
Llitjos	May 2020	France	retrospective, multicenter	yes	26
Lodigiani	April 2020	Italy	retrospective, single-center	informed consent	61
Longchamp	May 2020	Switzerland	prospective, observational, single-center	inf. consent waived	25
Maatman	May 2020	USA	retrospective, multicenter	inf. consent waived	109
Masi	June 2020	France	prospective; observational; single-center;control	informed consent	17
Mei	July 2020	China	retrospective, single-center, control	inf. consent waived	45
Middeldorp	May 2020	Netherlands	retrospective, single-center	inf. consent waived	75
Moll	July 2020	USA	retrospective; single-center	yes	102
Nahum	May 2020	France	prospective, observational, single-center	inf. consent waived	34
Nougier	July 2020	France	prospective, observational, single-center	informed consent	48
Patel	July 2020	UK	retrospective, single-center	inf. consent waived	39
Pavoni	May 2020	Italy	retrospective, single-center	inf. consent waived	40
Poissy	April 2020	France	retrospective, single-center, control	yes	107
Ren	July 2020	China	prospective, observational, multicenter	informed consent	48
Ranucci	May 2020	Italy	prospective, observational, single-center	yes	16
Shah	September 2020	UK	retrospective; multicenter	no	187
Siguret	July 2020	France	prospective, observational, single-center	yes	74
Soumagne	June 2020	France/Belgium	retrospective; multicenter	inf. consent waived	375
Spiezia	April 2020	Italy	observational, single-center	informed consent	22
Stessel	July 2020	Belgium	prospective; interventional; single-center	inf. consent waived	72
Tavazzi	April 2020	Italy	observational, single-center	informed consent	54
Thomas	April 2020	UK	retrospective, single-center	inf. consent waived	63
Torres-Machorro	September 2020	Mexico	prospective; observational; single-center	informed consent	30
van Veenendaal	September 2020	Netherlands	retrospective; single-center	NR	47
Voicu (a)	May 2020	France	prospective, observational, single-center	informed consent	56
Voicu (b)	September 2020	France	prospective; observational; single-center	yes	92
Whyte	July 2020	UK	retrospective, single-center	inf. consent waived	222
Yuriditsky	June 2020	USA	retrospective, single-center	inf. consent waived	64
Zangrillo	April 2020	Italy	retrospective, single-center	yes	73
Zermatten	August 2020	Switzerland	retrospective; single-center	informed consent	100

Supplementary Table S2: Risk of bias assessment

Table S2 depicts the detailed quality assessment for the 54 included studies.

First author	Question 1	Question 2	Question 3	Question 4	Question 5	Question 6a (PE)	Question 6b (DVT)	Question 7a (PE)	Question 7b (DVT)	Question 8a (PE)	Question 8b (DVT)	Question 9	Question 10	Question 11
Al-Samkari	1	1	1	1	1	1	1	1	1	0	0	0	1	moderate
Aleva	1	0	1	1	1	0	0	0	0	0	0	1	1	high
Avruscio	1	1	0	1	1	1	1	1	1	0	1	1	0	high
Azoulay	1	1	1	1	1	0	NR	0	NR	0	NR	0	0	high
Beun	1	0	1	1	1	0	0	1	1	0	0	0	1	high
Bilaloglu	1	1	1	1	0	0	0	0	0	0	0	0	1	high
Chen	1	0	0	1	1	NR	0	NR	1	NR	1	1	1	high
Chistolini	1	1	1	1	1	0	NR	0	NR	0	NR	0	0	high
Criel	1	1	1	1	1	NR	0	NR	1	NR	1	0	1	high
Cui	1	1	0	1	1	NR	0	NR	1	NR	0	0	1	high
Desborough	1	1	1	1	1	1	1	1	1	1	1	0	1	moderate
Devreese	1	1	1	1	1	NR	0	NR	1	NR	0	1	1	moderate
Dujardin	1	1	1	0	1	1	0	1	1	1	1	0	1	high
Elbadawi	1	1	1	1	1	NR	NR	NR	NR	NR	NR	0	1	high
Fraissé	1	1	1	1	1	0	0	0	0	0	0	0	1	high
Gibson	1	1	0	0	1	NR	0	NR	1	NR	1	0	1	high
Grandmaison	1	1	1	1	1	0	0	1	1	1	1	0	1	high
Hekimian	1	0	0	0	1	1	NR	1	NR	0	NR	0	0	high
Helms	1	1	0	1	1	1	0	1	0	1	0	0	1	moderate
Hippensteel	1	1	0	1	1	0	0	1	1	0	0	0	1	moderate
Kapoor	1	1	0	0	1	1	0	1	1	1	1	1	1	moderate
Klok	1	1	1	1	1	1	1	1	1	0	0	1	1	moderate
Lemos	1	1	0	0	1	0	0	0	0	0	0	0	1	high
Lendorf	1	1	1	1	1	0	NR	1	NR	0	NR	0	1	high
Llitjos	1	1	1	1	1	0	0	1	1	1	1	0	1	high
Lodigiani	1	1	1	0	1	0	1	1	1	1	1	0	0	high
Longchamp	1	0	1	1	1	1	1	1	1	1	1	0	1	moderate
Maatman	1	1	1	1	1	0	1	1	1	1	1	1	1	moderate
Masi	1	1	1	0	1	0	NR	0	NR	0	NR	0	0	high
Mei	1	1	1	1	1	0	0	0	0	1	1	0	1	high
Middeldorp	1	1	1	1	1	1	0	1	0	0	0	1	1	high
Moll	1	1	1	1	1	1	1	1	1	1	1	1	0	moderate
Nahum	1	1	1	1	1	NR	0	NR	1	NR	1	0	1	high
Nougier	1	1	0	0	1	0	0	0	0	0	0	0	1	high
Patel	1	0	0	1	1	0	0	1	1	1	0	0	1	high
Pavoni	1	1	1	1	1	0	0	1	1	0	1	0	1	moderate
Poissy	1	1	1	1	1	1	0	1	0	1	0	1	1	high
Ren	1	1	1	0	1	NR	1	NR	1	NR	1	1	1	high
Ranucci	1	0	0	1	1	0	0	0	0	0	0	0	0	high
Shah	1	1	1	1	1	0	0	1	1	1	1	1	1	high
Siguret	1	1	1	1	1	0	0	1	1	1	1	0	1	high
Soumagne	1	1	0	1	1	0	0	0	0	0	0	0	1	high
Spiezia	1	1	0	0	1	NR	0	NR	0	NR	0	0	1	high
Stessel	1	1	1	1	1	NR	0	NR	1	NR	1	0	0	high
Tavazzi	1	1	1	0	1	0	0	1	1	0	0	0	1	high
Thomas	1	1	1	1	1	1	1	1	1	1	1	0	1	moderate
Torres-Machorro	1	1	0	0	1	NR	0	NR	1	NR	1	0	1	high
Veenendal	1	1	0	0	1	1	1	0	0	1	1	1	1	high
Voicu (a)	1	1	0	0	1	NR	0	NR	1	NR	1	0	1	high
Voicu (b)	1	1	0	1	1	0	1	1	1	0	1	0	0	high
Whyte	1	0	0	1	1	1	NR	1	NR	1	NR	0	1	high
Yuriditsky	1	0	0	1	1	0	0	1	1	1	1	1	1	high
Zangrillo	1	0	1	1	1	0	NR	0	NR	0	NR	1	0	high
Zermatten	1	1	1	0	1	1	1	1	1	1	1	0	1	moderate

Supplementary Table S3: Patient characteristics

Table S3 shows relevant patient characteristics and type of pharmacological thromboprophylaxis/anticoagulation used. Data are presented as proportion or mean. BMI: body mass index; ICU LOS: intensive care unit length of stay; NR: not reported. Disease duration: duration of disease until ICU admission, *three studies reported the duration of disease until first venous thromboembolic event.

First author	Female	Age	BMI	Mortality	ICU-LOS	Disease duration	Malignancy	Anticoagulation
	%	years	kg/m^2	%	days	days	%	
Al-Samkari	6	65	NR	19	NR	NR	NR	any dose heparin
Aleva	34	65	29	26	17	NR	4	prophylactic heparin
Avruscio	20	67	28	NR	27	NR	12	other
Azoulay	23	62	28	26	10	8	NR	NR
Beun	NR	NR	NR	NR	NR	NR	NR	high-dose heparin
Bilaloglu	NR	NR	NR	54	NR	NR	NR	prophylactic heparin
Chen	39	63	NR	23	22	28*	6	prophylactic heparin
Chistolini	37	66	NR	NR	NR	NR	NR	any dose heparin
Criel	33	65	30	13	NR	NR	NR	any dose heparin
Cui	54	60	NR	10	NR	NR	NR	none
Desborough	27	59	28	30	9	NR	8	any dose heparin
Devreese	10	63	NR	13	25	NR	19	any dose heparin
Dujardin	23	62	27	NR	NR	NR	NR	any dose heparin
Elbadawi	NR	NR	NR	NR	NR	NR	NR	NR
Fraissé	21	61	30	41	NR	17*	NR	any dose heparin
Gibson	21	64	NR	24	NR	NR	NR	any dose heparin
Grandmaison	38	66	28	NR	NR	NR	7	prophylactic heparin
Hekimian	NR	NR	NR	NR	NR	NR	NR	NR
Helms	19	63	NR	9	10	NR	6	any dose heparin
Hippensteel	42	57	33	24	NR	NR	3	any dose heparin
Kapoor	42	60	30	47	7	NR	10	other
Klok	24	64	NR	22	NR	NR	3	any dose heparin
Lemos	20	57	34	20	NR	NR	0	any dose heparin
Lendorf	85	64	29	25	NR	12	10	any dose heparin
Llitjos	23	68	30	12	NR	7	0	any dose heparin
Lodigiani	20	61	NR	NR	12	NR	3	prophylactic heparin
Longchamp	36	68	28	20	NR	10	8	prophylactic heparin
Maatman	43	61	35	25	13	8	NR	prophylactic heparin
Masi	29	48	31	NR	NR	NR	NR	prophylactic heparin
Mei	NR	NR	NR	NR	NR	NR	NR	prophylactic heparin
Middeldorp	23	62	27	NR	15	NR	4	any dose heparin
Moll	42	65	30	27	15	NR	21	prophylactic heparin
Nahum	26	62	31	NR	NR	NR	3	prophylactic heparin
Nougier	NR	63	29	NR	NR	NR	NR	prophylactic heparin
Patel	18	53	31	NR	NR	NR	NR	any dose heparin
Pavoni	40	61	28	13	8	12	NR	prophylactic heparin
Poissy	NR	NR	NR	14	NR	NR	NR	prophylactic heparin
Ren	46	70	NR	31	NR	NR	NR	prophylactic heparin
Ranucci	6	61	26	44	NR	NR	NR	other
Shah	34	57	28	32	15	NR	8	any dose heparin
Siguret	NR	64	29	26	NR	10	NR	any dose heparin
Soumagne	23	64	30	36	NR	NR	12	other
Spiezia	9	67	30	NR	NR	NR	0	prophylactic heparin
Stessel	32	67	26	26	12	NR	NR	any dose heparin
Tavazzi	NR	NR	NR	NR	NR	NR	NR	prophylactic heparin
Thomas	30	NR	NR	16	8	NR	2	prophylactic heparin
Torres-Machorro	23	62	28	NR	NR	NR	NR	any dose heparin
van Veenendaal	23	63	29	15	18	NR	NR	prophylactic heparin
Voicu (a)	25	NR	NR	NR	NR	NR	NR	prophylactic heparin
Voicu (b)	28	62	28	59	NR	12*	0	prophylactic heparin
Whyte	NR	NR	NR	NR	NR	NR	NR	any dose heparin
Yuriditsky	28	64	NR	30	NR	NR	0	any dose heparin
Zangrillo	16	61	27	23	11	9	3	other
Zermatten	26	64	NR	28	11	NR	3	any dose heparin